

Telemetry, tracking, and command (TT&C)

11

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1 Introduction: Key factors to consider when designing TT&C for CubeSats

SmallSats and CubeSats are subject to very stringent constraints in mass, power, and cost. Hence, in designing the telecommunication system, it is especially important to be aware of the key factors that can affect the design. Specifically, these are the following:

- *Spacecraft data generation, buffering, and computational power of the avionics:* it is important to quantify as early as possible in the design the amount of data to transmit, since this can significantly affect the design. Sometimes the amount of data will be given to the communication system designer, while in some cases it will be necessary to compute it using the information available such as number of sensors, type of sensors, amount of information (bits) acquired over time, and number of pictures. When the amount of data is computed, it is important to analyze the computational capabilities of the onboard avionic and whether or not it can implement some form of compression. Processors for SmallSats and CubeSats [1] are currently capable of significant data processing on board with limited power consumption. Hence the designer should explore all these possibilities whenever the initial estimate of the amount of data to transmit appears to be high. Finally, it is important to quantify the buffering capacity of the avionics to compute the optimal data rate that should be as low as possible to minimize power consumption but sufficient to relay all the data.
- *Electrical power system:* many of the CubeSat radios can consume a significant amount of power, even if just powered in receiving mode. Hence the designer of the communication system will evaluate the required power for the radio and the transmission time required to download the data.
- *Ground stations availabilities and characteristics:* different types of ground stations are used to support CubeSat missions: university stations, commercial networks, and stations managed by space agencies. University stations are mostly operating in UHF/VHF and S band [1]. They are generally low cost, with very minor scheduling constraints, but they suffer from limited geographic diversities (they are generally in very few locations), covering only a small portion of the satellite orbit with very limited passes. The commercial alternatives and the stations supported by space agencies present different characteristics. They offer multiple frequency bands (UHF, L, S, X, Ku, and in certain cases, Ka) allowing the designer to trade across the different alternatives. They are generally more reliable, and they can offer higher gains and lower noise temperature, thanks to powerful cooling equipment that can result in a considerable improvement of the link performance. Disadvantages of

these kinds of stations are mainly cost and availability as they generally support multiple missions at the same time.

- *Range and pointing capabilities:* free-space loss scales quadratically with the distance. Hence, closing the link with adequate margin is challenging for CubeSats, especially on interplanetary trajectories and considering the limitations in volume, mass, and power that inevitably constrain the type of antennas or amplifiers that can be used. Another important issue for CubeSats is pointing capabilities especially for higher gain antennas.
- *Spectrum and licensing:* especially for low Earth orbit missions in which the link can be closed at higher data rates (in the order of Mbps), it is important to remember that the final bottleneck in the effective data rate will be determined by how much bandwidth the CubeSat is legally allowed to use. For example, NTIA restricts the bandwidth to be 5 MHz or less for S-band telecommunication systems and 10 MHz or less for X-band telecommunication systems.

2 Telecommunication system design

This section describes a typical process of modeling and designing a communication system for a SmallSat mission. The key aspect of the process is the fact that it is iterative: this is due to the peculiar constraints that distinguish SmallSats and CubeSats from other kind of missions. In fact, as mentioned previously, small satellites differ from other missions in the computational capabilities, power availability, ground station characteristics, and limited range and pointing capabilities. Hence, to optimize the design, it is necessary to consider all these factors while trying to maximize the data return. An overview of the approach is presented in the following block diagram. The following subsections are dedicated to the different aspects of the approach (Fig. 1).

2.1 Requirements

The first step in the communication system design approach is the analysis of the requirements to identify all the constraints that cannot be changed or that can be changed with corresponding cost increases or reduction of mission objectives. The requirements considered in this section are the most commonly experienced [2] in the design of the communication system for small satellites:

- *Data volume:* it is important to estimate early on in the design process the amount of data to be transmitted. The requirement impacts small satellite missions heavily especially when a lot of images are transmitted: in these cases, techniques of compression are considered. It is also important not to forget that engineering data need to be transmitted as well. Fig. 2 shows an overview of the spacecraft downlink data flow.
- *Ground station:* in some cases, the ground station is selected independently from the design of the communication system. In other cases a trade-off can be performed considering metrics such as station locations, availabilities, frequency compatibility with the spacecraft, geometry, and passes duration.
- *Orbit:* in many cases, small satellites are launched as secondary payloads, and the orbit is determined by the primary mission. In these cases the orbit cannot be changed, and as an additional challenge, the orbit is generally unknown for a long time during the design phase.

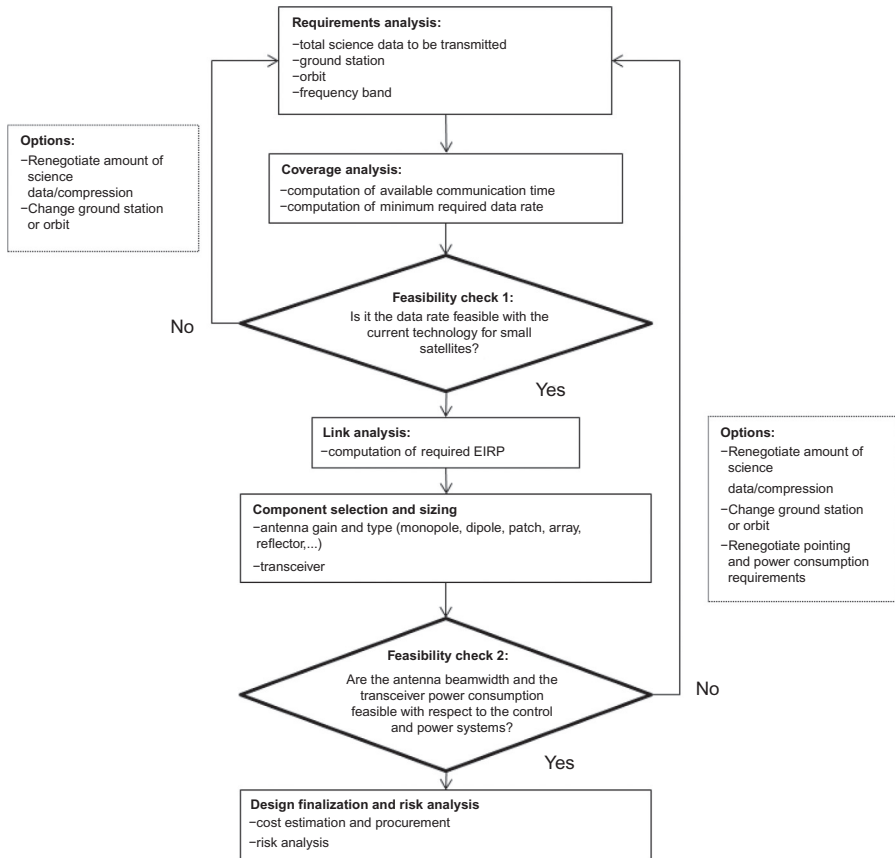


Fig. 1 Summary of the iterative process for designing a communication system for small satellites.

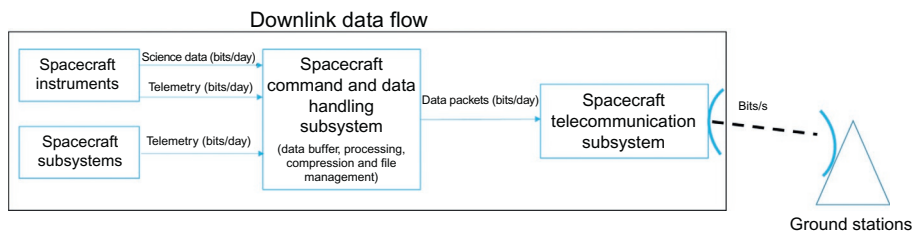


Fig. 2 Downlink data flow.

In some cases, altitude and inclination may be known; in some case, not even these data are available. The uncertainty of the orbit represents one of the biggest challenges for the designers of the communication system.

- *Frequency*: especially in the cases in which the ground station has already been selected, it is very common to face restrictions on which frequencies can be used. Additionally, constraints

on licensing can limit the choices in terms of band. Specifically, bands allocated for near-Earth spacecraft (any spacecraft at a distance of less than 2 million of kilometers from Earth) are different from those allocated for deep space spacecraft that can affect hardware selections.

When requirements are identified, a coverage and a link analyses for the mission can be performed.

2.2 Analysis

Two main analyses are performed when designing the telecommunication system: coverage and link. Coverage analysis is the first computational step in the design of the communication system, and it can be performed early on, as it depends mostly on ground station selection and orbit. Other factors can impact the coverage time, such as antenna directionality, pointing, ground station scheduling, and synchronization. However, for a first cut analysis of the communication system for small satellites, coverage can be considered a geometric problem assuming that ground station availability is not an issue; this is the case when a university ground station is used, especially when it is operating only one mission at the time. In that case, no conflicts with other missions or scheduling problems will arise, and the ground station will be able to communicate with the satellite whenever it is in visibility. If coverage is treated as a geometric problem, then simple orbital calculations can allow the designer to compute the coverage time and to compute the minimum required data rate. The steps are described in the succeeding text.

- Set up an orbital simulation: this step can be accomplished by using a commercial software [3] or by writing a simple code that uses basic astrodynamics equations [1]. The simulation should include the orbit of the satellite and the location of the ground stations. It should be noted that different orbits can provide different telecommunication characteristics. For example, a geostationary satellite positioned directly over the country of Ecuador will give users in that country 100% temporal coverage. Differently a polar-orbiting spacecraft will allow for good coverage around the polar regions, while worst coverage will be around the equator. Hence, if a particular orbit is not already selected, the optimal orbital characteristics for a particular satellite should be studied prior to designing the telecommunication system for the mission.
- Compute the coverage profile: the coverage profile is binary graph that indicates for any instant of time whether the satellite is in coverage or not. The mathematical formulation is

$$c(t) = \begin{cases} 1 & \text{if sat is covered} \\ 0 & \text{if sat is not covered} \end{cases} \quad (1)$$

- Create a data cumulating function: in prime approximation, it can be a linear function. If we define D as the total amount of data to be transmitted per day, the amount of data cumulated in function of time (bit/s) can be approximated by

$$f(t) = \frac{D}{86,400} \cdot t \quad (2)$$

- Compute the amount of data download over time to determine the minimum required data rate: in this last step the coverage profile and the data cumulating function are used to identify how much data rate is required to transmit the required amount of data. The approach is iterative. The algorithm starts with an initial data rate $r(t)$, which is increased or decreased depending on the result of the computation. The algorithm works as follows: for any instant of time, if the coverage function is zero, data are cumulated in the buffer ($b(t) = b(t-1) + f(t)$). If coverage is one, data are cumulated, but they are also transmitted at a speed given by the data rate ($b(t) = b(t-1) + f(t) - r(t)$).

The algorithm produces a triangular shaped plot where the data are cumulated for every instant of time in which there is no coverage (increasing ramp), while data are downloaded for every instant in which there is coverage. This plot provides the following information:

- (a) The peak of the triangles indicates how much data will be cumulated over time before a download, which helps to size the buffer memory required for the avionics.
- (b) Every time the amount of data stored reaches zero, it means that every data collected have been successfully downloaded. It is important to ensure that the data storage plot goes periodically to zero, otherwise the data rate $r(t)$ needs to be increased, and/or the data cumulation $f(t)$ needs to be decreased to ensure a complete download of mission data.

Link analysis follows coverage analysis in the design. It uses the data on ground station, orbit, and data rate to determine the minimum EIRP for the SmallSat. Link analysis needs to be computed for both downlink and uplink. However, it is generally recommended to start from the downlink that is key to estimate the kind of communication equipment that will need to be carried by the satellite. The steps are as follows:

- Computation of the quality requirement $\left(\left(\frac{E_b}{N_0} \right)_{\text{required}} \right)$: depending on which type of modulation is selected and on the bit error rate (BER) probability desired, a minimum E_b/N_0 can be computed [1].
- Computation of the ground station gain: the gain can be obtained directly by contacting the agency/company/team responsible for the stations, or it can be computed for a parabolic dish as (D is the diameter, η is the efficiency, and λ is the wavelength):

$$G_r = \eta \left(\frac{\pi \cdot D}{\lambda} \right)^2 \quad (3)$$

- Computation of the noise temperature: it is the sum of two components, the antenna noise and the internal (receiver) noise. The antenna's noise temperature is the thermal noise that originates from objects within the field of view of the antenna. The formula (see Ref. [1]) is the following:

$$T_A = \frac{1}{4 \cdot \pi} \iint T(\theta, \varphi) \cdot D(\theta, \varphi) \cdot \sin(\theta) \cdot d\theta \cdot d\varphi \quad (4)$$

However, since it can be difficult to identify the distribution of the noise temperature across the space ($T(\theta, \varphi)$), the antenna noise temperature is often measured

experimentally. Data on the noise temperature for the Deep Space Network (DSN) ground stations can be found in Ref. [4]. The internal (receiver) temperature depends mostly on the technology of the receiver. Generally, receivers are characterized by the noise figure NF :

$$T_r = 290 \cdot \left(10^{\frac{NF}{10}} - 1 \right) \quad (5)$$

- Computation of free-space loss at maximum distance (d measured in the same units as λ) as:

$$L_{d_{\max}} = - \left(\frac{4 \cdot \pi \cdot d}{\lambda} \right)^2 \quad (6)$$

- Computation of additional losses: a nonexhaustive list of these losses is as follows:
 - (a) Pointing losses experienced as a result of not precise pointing: Pointing errors can be static due to misalignments or variables. They can be approximately computed as:

$$L_p = -12 \cdot \left(\frac{e}{\theta} \right)^2 \quad (7)$$

where e is the pointing offset in degree and θ is the antenna beam width in degree.

- (b) Atmospheric losses (L_a): ionospheric losses (generally insignificant above 1 GHz and very little impact in the UHF Band [11]) and tropospheric losses due to rain, oxygen, and water vapor (they are negligible for UHF frequency, and they cause fractions of dB of attenuations in the S band. Tropospheric losses start causing significant attenuation for frequencies above 10 GHz). A summary of tropospheric losses can be found in Fig. 3. The reader can also find additional information in Ref. [5].
- (c) Line loss: caused by cables. If the length of all the cables is known (l) and also the coefficient of attenuation, then line loss results:

$$L_l = -\alpha \cdot l \quad (8)$$

- (d) Modem implementation losses (spectral efficiency): these depend on modulation, coding, and roll-off. The most basic modulation scheme is binary phase-shift keying (BPSK): information is coded using two phase states per symbol that corresponds to 1 bit of information per symbol. Higher-order modulations are quadrature phase-shift keying (QPSK), 8 phase-shift keying (8-PSK), 16-PSK, and more. The advantage of higher-order modulation schemes lies in their spectral efficiency, since they can encode more bits per symbol (QPSK encodes 2 bits per symbol, 8-PSK 3 bits per symbol, and 16-PSK 4 bits per symbol). However, as the modulation order increases, the sensitivity to phase-induced errors also increases. A key parameter in the modulation is the number of bits per symbol for the specific encoding scheme used (m). In addition to modulation, the use of forward error correction (FEC) coding is the idea of adding redundant bits that can be used to improve the ability of the receiver to correct bit errors. The redundant bits not only improve reception and correction but also can affect spectral efficiency. The coding efficiency can be computed as [1]:

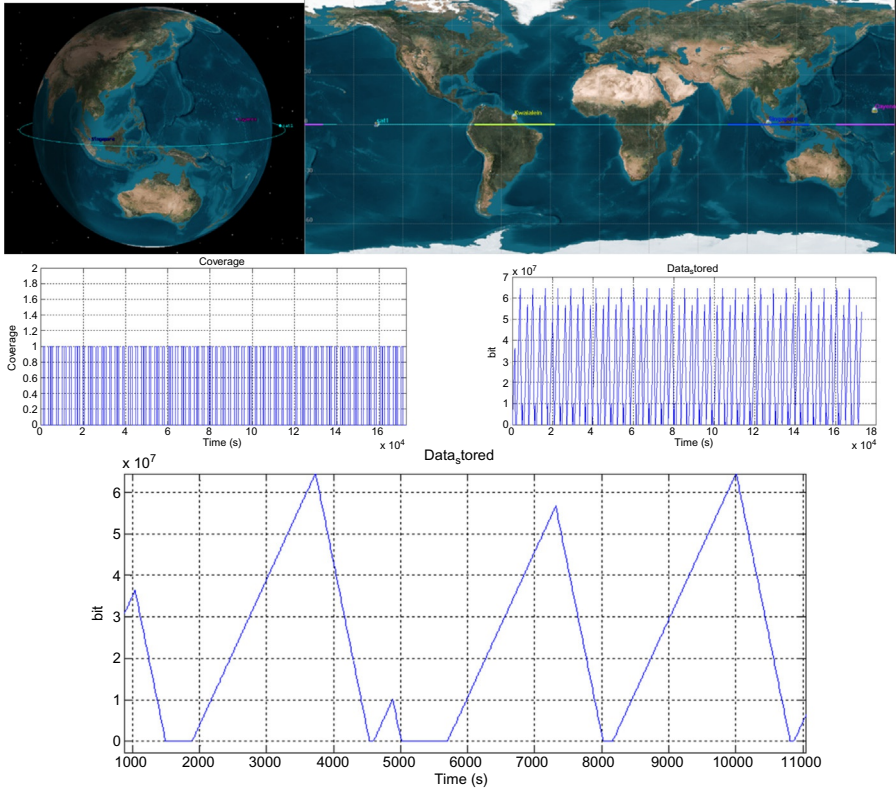


Fig. 3 Tropospheric attenuation [5].

$$k = \frac{n}{n+r} \quad (9)$$

where n is the number of information bits and r is the number of redundant bits added as a result of the specific coding scheme adopted. Roll-off factor (ρ) is the amplitude of the filters used to minimize interference across adjacent frequency bands. Spectral efficiency is [1]:

$$L_e = -10 \cdot \log_{10} \left(\frac{m}{k \cdot (1 + \rho)} \right) \quad (10)$$

- Computation of the minimum required EIRP: when all the previous elements have been computed, it is possible to calculate the minimum required EIRP, using the following formula:

$$EIRP_{\min} = \frac{E_b}{N_{0\text{Required}}} + M - L_{d_{\max}} - L_p - L_a - L_l - L_e - G_r - 228.6 + 10 \cdot \log(T_s) + 10 \log(R_{\min}) \quad (11)$$

The formula provides the EIRP in dBW as described in Ref. [1]. $R_{\min}$ indicates the minimum data rate, and M is the margin on the communication system, which is generally (rule of thumb) at least 3 dB.

2.3 Components selection and design finalization

Component selection is the process in which the designer selects COTS or custom-made components to reach the EIRP estimated through link analysis. Specifically the designer needs to pick at least an antenna and a transceiver (amplifiers are sometimes incorporated in the radio assembly). Additional antennas and transceivers for redundancy are generally not used for SmallSats and CubeSats, due to the volume and mass constraints. More information on components are included in the next section.

The last step in the design of the communication system is to finalize the design by performing cost estimation and risk analysis. The cost estimation can be basically broken into two main categories: components and labor. The estimation of the hardware cost can be easily performed when products are available on the market. More complex is the case in which the component needs to be fabricated. Labor cost is challenging to estimate, and it varies widely depending on the organization performing the work. Data on previous missions should be used if possible, to refine the estimates. A detailed analysis of cost modeling is beyond the scope of this chapter. The reader can find more in numerous reference documents; a good reference to start with is the new Space Mission Analysis and Design [1].

Regarding risk analysis, it is the process of estimating the risk that a given design will exceed mass and power consumption constraints over the temporal evolution of the design. Risk analysis [6] can be performed using two types of statistics: data and expert. To develop data statistics, it is necessary to create a database of already developed components. Until a few years ago, this was a great challenge in the field of small satellites, because very few components existed. However, in recent years, the development of SmallSats and CubeSats caused a great development of new products and components. Regarding expert statistics, it is important to take into account that SmallSats and CubeSats are still new in the engineering community. Hence, it is necessary to pay attention in selecting the expert statistics to reference to avoid over-estimation issues caused by engineers that are accustomed to work on larger spacecraft.

3 Telecommunication components for CubeSats

This section provides an overview of the main telecommunication components for CubeSats. The components presented are not meant to be an exhaustive list as this is an active field of research and new components are continuously being developed.

3.1 Antennas

The most common CubeSat antennas are low gain: monopole, dipoles, and patches [7]. The main characteristics of the antennas are summarized in Table 1.

For interplanetary missions, medium-gain antennas (MGA) and high-gain antennas (HGA) are in most cases necessary, due to the longer distances and higher free-space losses. MGA and HGA tend to be selected on CubeSats, which are at least six CubeSat units as the volume and side surfaces allow for their accommodations. MGAs for CubeSats are typically at X band, and they are generally designed as arrays of patches, which can fit on one of the 20 × 30 cm sides [8]. Typical MGAs are 4 × 4 and 8 × 8 elements. CubeSats HGA are generally at X band or Ka band, and they are mostly reflectarrays, mesh reflectors, or inflatables [8]. Characteristics are summarized in Tables 2 and 3 for MGA and HGA, respectively.

3.2 Radios

Commercial transceivers are currently available for SmallSats and CubeSats from a variety of vendors. These products mostly operate at UHF and S band, and they support data rates that range from a few kilobits per second to a few megabits per second. The main characteristics of some of these transceivers are summarized in Table 4 (NA indicates that data are unavailable). It is important to notice that depending on the type of mission (deep space vs near Earth), the frequency regulations limit the frequency

Table 1 Typical low-gain antennas for SmallSats and CubeSats.

Antenna	Typical band of operation	Typical gain (dBi)	Typical half power beamwidth	Polarization	Typical volume	Typical mass (g)
Monopole	Mostly UHF/VHF It can be used also in S band	0–3	Omni	Linear or circular	10 mm (diam) × $\lambda/4$	<20
Dipole	Mostly UHF/VHF It can be used also in S band	0–3	Omni	Linear or circular	10 mm (diam) × $\lambda/2$	<30
Patch	S band and X band	5–8	60 degrees	Linear or circular	10 × 10 × 0.1 cm (S band) 5 × 5 × 0.1 cm (X band)	<50

Table 2 Typical medium-gain antennas for CubeSats.

MGA	Typical band of operation	Typical gain (dBi)	Typical half power beamwidth (degrees)	Typical volume	Typical mass (g)
4 × 4	X band	11	~26	10 × 10 × 0.1 cm	<50
8 × 8	X band	23	~7	20 × 20 × 0.1 cm	<200

Table 3 Typical high-gain antennas for CubeSats.

HGA	Typical band of operation	Typical gain	Typical half power beamwidth	Typical volume	Typical mass (kg)
Reflectarray	X band and Ka band	29.2 dBi for 60 × 30 cm array at X band [8] 26 dBi for 30 × 30 cm array at Ka band [8]	~3 degrees (X band) ~0.5 degrees (Ka band)	60 × 30 × 0.5 cm at X band (once deployed) 30 × 30 × 0.5 cm at Ka band (deployed)	<1
Inflatable	X band and Ka band	29 dBi for 0.7 m reflector at X band [9]	~3 degrees (X band)	10 × 10 × 5 cm (stowed)	<1
Mesh reflectors	X band and Ka band	42 dBi for a 0.5 m deployable at Ka band [8]	~0.7 degrees (Ka band)	1.5 U (stowed)	<2

bands that can be used. As a result, not all the transceivers shown in Table 4 can be used for every mission. The reader should always verify that a given transceiver can be compliant with the particular frequency regulations required for any given mission.

For deep space applications where a transponder is needed, the main radio currently used is the Iris radio developed at the Jet Propulsion Laboratory. Designed for both near-Earth and Deep Space X-band frequency allocations, the Iris radio provides telecommunication and navigation services for CubeSat missions. Originally developed for the Interplanetary NanoSpacecraft Pathfinder in Relevant Environment (INSPIRE) program [20], the Iris design was subsequently improved for use on the Mars Cube One (MarCO) mission. MarCO-A and MarCO-B carried the Iris radio, and they were successful in relaying back to Earth the data for the InSight Mars entry, descent, and landing (EDL) [21]. The new version of the Iris radio, developed after

Table 4 Transceivers for SmallSats and CubeSats [10–19].

Component	Manufacturer	Transmitting power	Supported data rate	Frequency band	Modulations and coding	Dimensions	Mass (g)
Beryllium S-band transceiver	AstroDev	2 mW	NA	2400–2485 MHz (up and down)	GFSK	$46 \times 46 \times 10.5$ mm	32
Lithium 1 radio	AstroDev	250 mW–4 W	9.6, 38.4, 76.8 kbps	130 (up)–450 MHz (down)	FSK-GMSK	$10 \times 33 \times 65$ mm	52
S-band transmitter	Clyde Space	6 W max	2 Mbps (max)	2200–2300 MHz	QPSK-OQPSK	$96 \times 90 \times 16$ mm	80
UHF-VHF transmitter	Clyde Space	4–10 W	9.6 kbps	2400–2483 MHz 420–450 MHz 130–150 MHz	GMSK-AFSK	$96 \times 90 \times 16$ mm	90
NanoCom U482C	GomSpace	1 W	1.2, 4.8, 9.6 kbps	435–438 MHz (up and down)	MSK	$95.4 \times 90.15 \times 18$ mm	80
VHF-UHF transceiver	ISIS	1.7 W	1.2, 2.4, 4.8, 9.6 kbps	400–450 MHz (down) 130–160 MHz (up)	BPSK, AFSK	$96 \times 90 \times 15$ mm	85
S-band transceiver	ISIS	3.5 W	100 kbps	2100–2500 MHz (up and down)	BPSK, GSMK	$90 \times 96 \times 15$ mm	62
Nano 2420	Microhard	1 W	115–230 kbps	2400–2483 MHz (up and down)	BPSK	$32 \times 51 \times 6.35$ mm	19
Nano 920	Microhard	1 W	1.2 Mbps	920–928 MHz (up and down)	BPSK	$32 \times 51 \times 6.35$ mm	19
SWIFT-XTS	Tethers Unlimited	1–7 W	Up to 100 Mbps	8400–8450 MHz (down) 2100–2400 MHz (up)	BPSK, OQPSK	$86 \times 86 \times 50$ mm	NA

Continued

Table 4 Continued

Component	Manufacturer	Transmitting power	Supported data rate	Frequency band	Modulations and coding	Dimensions	Mass (g)
CSR-SDR-S/S	Vulcan Wireless	1–2 W	1 Mbps	2200–2400 MHz (down) 2025–2110 MHz (up)	OQPSK	85 × 85 × 40 mm	380
SCR-100	Innoflight	3 W	Up to 4.5 Mbps	2200–2400 MHz (down) 2025–2110 MHz (up)	BPSK, QPSK, GMSK	82 × 82 × 45 mm	330
EWC27 + OPT27-SRX	Syrlinks	2 W	Up to 100 Mbps	8400–8500 MHz (down) 2025–2110 MHz (up)	BPSK, OQPSK	85 × 85 × 50 mm	700
μSDR-C	Space Micro	5 W	Up to 42 Mbps	70 MHz–3 GHz	BPSK, QPSK, 8-PSK	102 × 117 × 100 mm	750

MarCO, is targeted to serve 6 of the 13 secondary CubeSat missions that will act as secondary payloads onboard the NASA Space Launch System (SLS) Exploration Mission 1 (Artemis 1): BioSentinel, Lunar IceCube, Lunar Flashlight, LunaH-Map, Near-Earth Asteroid Scout (NEA Scout), and CubeSat for Solar Particles (CuSP). The Iris radio is designed to operate at X band, in coherent and noncoherent mode with 880/749 turn around ratio and less than 5-dB noise figure. The uplink modulation is PCM/PSK/PM with BCH encoding and a variety of data rates that are compatible with the NASA Deep Space Network. The downlink modulation is BPSK with several encoding options (Manchester, suppressed carrier, and subcarrier) and coding schemes (Reed-Solomon, convolutional, and turbo). Downlink data rates range from 62.5 bps to 256 kbps. For navigation functions the Iris radio provides turnaround ranging and Delta-Differential One-Way Ranging (Delta-DOR) tones. Different from other CubeSat radios where generally the low-noise amplifier (LNA) and solid-state power amplifier (SSPA) are included in the radio chassis, the Iris radio requires an external LNA and SSPA. The Iris LNA provides two ports for up to two receiving antennas. The SSPA provides three input ports for the connection of up to three transmitting antennas. The RF output power of the SSPA comes in two options: 2 or 4 W. The Iris transponder is an enabling technology that made the MarCO mission successful and makes the Artemis 1 missions possible.

4 Optical telecommunications for CubeSat

Optical communication systems for CubeSats are currently under study. They have the potential of enabling compact and low power uplink/downlink for SmallSat interplanetary missions. A paper from Staehle [22] describes a possible optical communication package for CubeSat to occupy 1U in a 6U mission concept, designed for interplanetary CubeSat exploration. The proposed package is based on JPL laser telecommunication developments, and it is supposed to achieve a link capacity of 1–4 kbps at 2AU of distance from Earth. The future use of optical communication for CubeSats and small spacecraft is a very promising option. However, some challenges need to be addressed before being able to actually develop an optical communication system for small platforms:

- Packaging: more work needs to be done to incorporate a flight laser communication payload into a 1U CubeSat. Specifically the terminal optical assembly needs to be reduced up to a point where it can really fit the CubeSat standard dimensions. In Ref. [22], different assembly options are mentioned.
- Transmitting power (laser): in Ref. [22] a 10% efficiency, highly compact, master oscillator laser is proposed. However, improvements of the laser efficiency are required to increase its transmitting power up to 5 W, which is what is necessary in Ref. [22] to close the interplanetary link at 2AU.
- Control and pointing: microradian-level fine pointing is required to successfully communicate using lasers. Multiple efforts are currently being conducted worldwide to develop fine control and pointing for miniaturized platforms. However, it is a very challenging problem,

and microradian-level fine pointing for small spacecraft has not been yet successfully demonstrated in space.

If these challenges are successfully addressed, optical communication will greatly improve the interplanetary capabilities of small spacecraft.

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